

ESSA JAN

+1 (401) 644-7735 | essajankhattak@gmail.com | github.com/essacheez | linkedin.com/in/essa-jan/

EDUCATION

Brown University

Master's, Computer Science | GPA: 4.0

Aug 2025 - May 2027 expected

Providence, RI

Relevant coursework. Reinforcement Learning, Self Supervised Learning and World Models, Learning with Limited Data, AI Safety and Alignment.

Lahore University of Management Sciences

Bachelor's, Computer Science | GPA: 3.74

Sep 2021 - May 2025

Lahore, Pakistan

Merit Scholarship 2022-2023. Dean's Honour List 2021-2025.

EXPERIENCE

Preference Model

RL Engineering Intern

May 2026 - Aug 2026

- Improved KernelBench performance from 0.23x to 0.80x with trajectory correction that identified failure points and added one or two corrective turns from a stronger model.
- Built 200+ RL tasks for LLM post-training and parameter-efficient development. Integrated environment infrastructure with training rollouts and adapted five datasets and benchmarks to in-house harnesses.

Brown University

Research Assistant

Providence, RI

Sep 2025 - Apr 2026

- Built an end-to-end RL data pipeline that generated 3,200 facts and 22,000 QA pairs through automated scraping, fact extraction, and LLM question generation.
- Implemented distributed GRPO training across three GPUs with TRL and cut training time by about 25% by using vLLM for batched inference and reward computation.

Technology for People Initiative, LUMS

AI Engineer

Lahore, Pakistan

May 2025 - Aug 2025

- Built a multi-agent tax assistant for B2B and B2C processing. Added FastAPI endpoints for PDF, CSV, Excel, and image notices using Pydantic schemas, OCR, and table extraction.
- Developed a RAG pipeline for six file formats, background question generation, document verification, and Section 161 mismatch handling. Made LLM name matching robust to malformed JSON and multi-document inputs.

SELECTED PROJECTS

ShardCompute | Distributed LLM Inference

Python, MLX, asyncio, WebSockets, FastAPI

- Contributed 2,310 lines across five commits to a multi-Mac transformer runtime. Replaced the TCP peer mesh with a coordinator WebSocket relay using a 16-byte binary header, async receive queues, locks, and timeouts.
- Added pipeline parallel execution and model sharding support. Diagnosed 4-bit MLX weight layouts and expanded collective tests for all-reduce and all-gather paths.

Post-training Coding Evaluation Harness

Python, TRL, PEFT, vLLM, uv

- Built a command-line evaluation harness with a pluggable benchmark registry for HumanEval, HumanEval+, MBPP, and BigCodeBench. It discovers checkpoints and runs pass@1 evaluation through a vLLM server with sandboxed code execution.
- Added persona metrics for style similarity, phrase overlap, and marker rates to measure the capability and behavior trade-offs of a LoRA adapter.

Hospital Management System

React, TypeScript, Express, MongoDB

- Built role-specific patient, doctor, and staff dashboards with separate sign-up and login flows. Implemented appointment scheduling, prescriptions, payments, feedback, and cross-role search.
- Designed an Express and Mongoose backend with 15 REST endpoints and five core schemas for doctors, patients, staff, appointments, prescriptions, and feedback.

SKILLS

Languages. Python, TypeScript, JavaScript, Java, C++, SQL, Bash, Haskell.

Backend and systems. FastAPI, Node.js, Express, REST APIs, WebSockets, asyncio, Pydantic, MongoDB, Mongoose, SQLite, Docker, Linux.

Frontend and tools. React, Vite, Tailwind CSS, Git, pytest, Jest, Slurm, vLLM, MLX.